Yuhang (Olric) Zeng

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Education

Cornell University Ithaca, NY

Bachelor of Science in Computer Science, GPA: 4.11/4.00 *Expected May 2028*

- Coursework: Data Structures and OOP, Backend Development, Databases, Computer System Organization, Mathematical Foundations of Computing, Functional Programming (OCaml), Linear Algebra

Experience

Hack4Impact Cornell Ithaca, NY

Software Engineer - Mavuno *Nov. 2025 – Present*

- Designed and implemented an Express and PostgreSQL announcement service for Mavuno's crop-support platform, choosing persistent inboxes over push-only delivery to preserve crop guidance through connectivity loss, with asynchronous delivery and retry tracking for thousands of farming households
- Built targeted broadcasts across **3 regions** and **2 languages**, processing recipients in batches to limit memory usage and enforcing database transactions and uniqueness constraints to prevent duplicate announcements
- Reviewed pull requests and tested APIs, collaborating with developers and designers on a 10-member team.

Sitefit AI Remote

Software Engineering Intern *Jun. 2026 – Aug. 2026*

- Implemented a TypeScript site-planning engine that turns public ArcGIS parcel data into editable commercial layouts, supporting multi-parcel and master-planning workflows for large commercial developments
- Reduced parking-layout generation from **9.7 min** to ∼**2.2 s** on a 69-acre parcel by replacing exhaustive search with a bounded solver targeting 1,150 parking spaces across a 608-vertex site boundary
- Moved PDF, DXF, IFC, and DAE exports off the API request thread into asynchronous jobs persisted in PostgreSQL, returning status immediately and cutting p95 response time from **74 ms to 0.06 ms**

Cornell Ezra Systems Research Group Ithaca, NY

Student Researcher & Developer *Jan. 2026 – May 2026*

- Created HTML/CSS interfaces and a real-time Node.js API pipeline that integrates weather services, normalizes live metrics across **five** cities, and serves data through Next.js routes for published urban-modeling research
- Architected a sun/shade routing engine for walking and cycling; refactored **800+** lines, replacing modulo-based marker placement with coordinate normalization to fix overlap and cut map re-render time by **30%**

Coding Mind Irvine, CA

Software Engineering Intern *Jun. 2024 – Jul. 2024*

- Built an Android nutrition app with Flutter, Firebase, and TensorFlow Lite for personalized dietary guidance
- Designed and deployed Python/Flask REST APIs to process nutrition data and deliver model predictions
- Tuned model parameters to improve prediction accuracy from **40% to 66%** using a dataset of **101,000+** images

Projects

MindDo AI Student Showcase | *FastAPI, Celery, Redis, Kubernetes, KEDA, AWS, Playwright* 2026

- Engineered an automated pipeline for MindDo.ai that converts any website URL into a narrated demo video, AI-written summary, printable flyer, and QR code using Playwright, Claude Sonnet, OpenAI TTS, and ffmpeg
- Deployed an AWS (EC2, S3) pipeline on k3s with Celery workers autoscaled by KEDA on Redis queue depth; a **20**-job load test scaled **1 to 4** workers in **17 s** and completed **20/20** jobs with zero job loss
- Cut repeat-submission latency ∼**99%** (**30 s** to **0.14 s**) with a content-addressed cache keyed on page-content hash

Luminary - Study Tracker App | *Flask, SQLAlchemy, Docker, GCP* 2025

- Led backend development for a **5-person** team, building **25+** REST endpoints with per-minute user progress sync
- Designed the multi-user session schema, containerized with Docker, and shipped a live GCP deployment
- Shipped an OpenAPI spec and won **Best Overall App out of 12 teams** at the 2025 AppDev Hack Challenge

Technical Skills

Languages: JavaScript, HTML/CSS, SQL, Python, TypeScript, Java, OCaml
Backend & APIs: RESTful APIs, Node.js, NoSQL, GraphQL, FastAPI, Flask, SQLAlchemy
Infrastructure & Tools: Git, Linux/Unix, Docker, AWS, PostgreSQL, MySQL, Postman, Redis, MongoDB, GitHub Actions CI/CD, Kubernetes, GCP, Firebase, Supabase, Prisma